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Topic : Docker File : Explanation :

Based on the code : in Docker file :

COPY requirements.txt .

Question :

- Where and How it copy ?
- If local computer which folder it copies ?
- How many files will copy ?

Concepts :

- Basically take requirements.txt from my local computer and copy it into the containers current folder
- COPY is the docker command to copy files and requirements.txt is the source file on your computer and (.) is the destination inside the docker container
- This is like moving a specific file from your local computer into a docker image
- This is like a bridge between two completely different computers such as local machine which is the Host and the docker container which is the guest

What does COPY requirements.txt . mean ? :

- This is has 2 parts such as requirements.txt which is the source from your local machine and (.) is the destination (current folder inside container)
- This command copies one specific file from your local computer into the docker container
- In docker file , this command is used to transfer a specific file from you local machine into a docker image during the build process

From where in local system is the file copied ? :

- Docker copies from your build context (structure) and it is the folder where you run the command : docker build .
- Based on our structure , If you run : cd app1 and docker build .
- This goes inside the app1 folder and docker will copy : app1/requirements.txt which is the container
- Also docker looks for requirements.txt in the build context and running the folder as docker build .
- If you run command inside : myproject/app1 then the docker will look for : myproject/app1/requirements.txt
- This copies from the exact folder where your run the build command.
- If your docker file is in myproject/app1/ and you are building app1 , it will look for requirements.txt inside that app1 folder

Where inside the container is it copied ? : Where and how does it copy ? :

- We have this command : WORKDIR /app , this means current working folder inside container which is /app
- So COPY requirements.txt . becomes /app/requirements.txt
- This command : WORKDIR /app , means docker creates a folder /app inside the container and all next commands runs inside /app
- So COPY requirements.txt . becomes COPY requirements.txt /app

How many files are copied ? :

So only one file is copied because COPY requirements.txt . Actually copies only requirements.txt

This only copies the single file named requirements.txt

In Simplified Way :

We have local folder as app1 and container folder as /app

Now we use COPY requirements.txt . which means take only requirements.txt from my local folder such as app1 and put it inside the container folder which is /app

Based on our project structure :

If you build inside app1 : cd app1 and docker build .

Then using COPY requirements.txt . which copies from app1 and not app2

Therefore , Summary : COPY requirements.txt .

- Can copy only the file requirements.txt from my local folder app1 into the container folder /app

So where is it copied from ? :

- From your current folder and if you run cd app1 and docker build .
- Docker takes the file from app1/requirements.txt
- Also looks at the folder on your local computer where docker file is sitting in myproject/app1/ and it finds the file name requirements.txt
- Therefore computer folder as : myproject/app1/requirements.txt

So where is it copied to ? :

- Inside container in : /app/requirements.txt , because of WORKDIR /app
- Docker has already created a folder called /app and the (.) dot at the end of the command is for right here in the current folder
- So the file lands at : /app/requirements.txt. which is inside the container

How many files are copied ? :

- Only One file such as requirements.txt

So therefore ,

- We pick only requirements.txt from app1 folder and put inside the container folder such as/app
- Copies 1 file only (requirements.txt) from your current project folder app1 to /app inside container folder
- COPY requirements.txt . means you copy one file into /app inside container